

Trainings ▾

Leak Test

WARNING

The external pressure tank used must meet SAE J10 and FMVSS121 standards, and have a safety pressure relief valve which opens between [150 to 175 psi]. Failure to use the proper pressure vessel and plumbing can result in property damage and serious personal injury.

Air Compressor Diagnostic Test

- Park the vehicle. Use wheel chocks or an appropriate anti-roll device to stabilize the vehicle.
- Drain the vehicle air system.
- Remove the air discharge hose and air governor signal hose from the air compressor. Install pipe plugs into the air compressor unloader signal ports.
- Plumb the air compressor discharge hose into an external pressure tank. The external pressure tank **must** be equipped with a pressure gauge and pressure relief valve. Make sure that the fittings are install with appropriate thread sealant and do **not** leak.

Start the engine and operate at rated engine speed.

Verify that the air compressor will build pressure in the external tank. If air pressure fails to build in the pressure tank, the air compressor has malfunctioned. Refer to Procedure 012-013 in Section 12. (</qs3/pubsys2/xml/en/procedures/100/100-012-013.html>)

Note : Once the external pressure tank pressure reaches 689 kPa [100 PSI], shut the engine down.

Monitor the pressure gauge on the external tank for rapid leak down of the air pressure. If the air tank loses more than 138 kPa [20 PSI] in a 5 minute period, the pressure relief valve, intake valve, or exhaust valve has malfunctioned. The air compressor cylinder head intake and exhaust valves are **not** serviceable. Replace the air compressor cylinder head. Refer to Procedure 012-007 in Section 12. (</qs3/pubsys2/xml/en/procedures/100/100-012-007.html>)

Note : Do not cycle air accessories such as seats, doors, wipers, air bags, etc.

- If there is any noticeable decrease of the air gauge readings or the air dryer cycled during the 10 minute test, repair the leaks

Note : Leaks in systems that hold pressure for 5 to 10 minutes may be hard to find.

Preparatory Steps

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

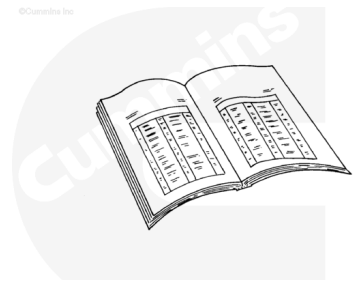
Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Disconnect the batteries. Refer to the OEM service manual.
- Steam clean the air compressor.
- Drain the engine cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Refer to the OEM service manual or remove the hydraulic pump, if equipped.

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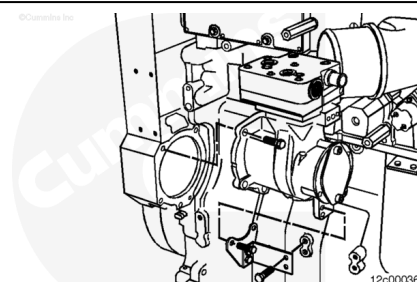
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- Remove any attachments on the rear of the air compressor. Refer to the OEM service manual.
- Remove the coolant lines from the air compressor.
- Remove the air connections from the air compressor.
- Remove the air inlet and air outlet connections from the air compressor.

Remove

Remove the air compressor support brackets and capscrews.

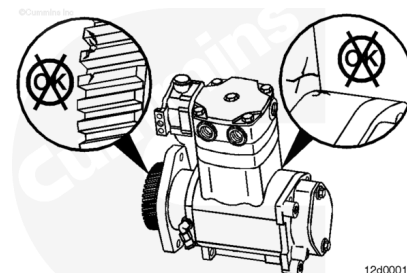
Remove the two capscrews and the air compressor.



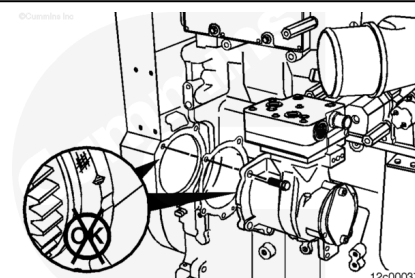
Clean and Inspect for Reuse

Inspect the compressor housing for cracks or other damage.

Inspect the drive gear for cracks or other damage.

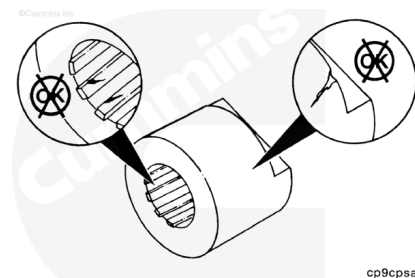


Make sure the gasket surfaces of the front gear housing and air compressor are clean and **not** damaged.



Inspect the air compressor adapter at the rear of the air compressor.

Replace the adapter, if worn. Use air compressor adapter replacer, Part Number 3165101, for 11-tooth adapters or air compressor adapter replacer, Part Number 3165102, for 13-tooth adapters.

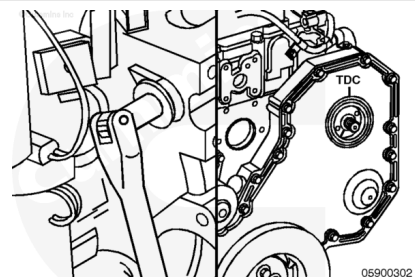


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Install

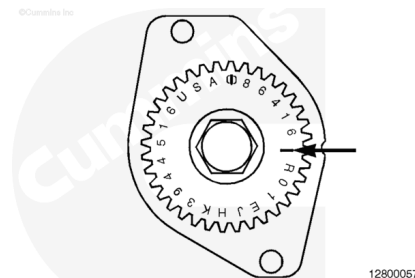
Air Compressor Timing (for Single-Cylinder Air Compressor Only).

Rotate the engine so that the number 1 cylinder is at top dead center (TDC) on the compression stroke. This is done by aligning the timing mark on the fuel pump gear with the TDC mark.



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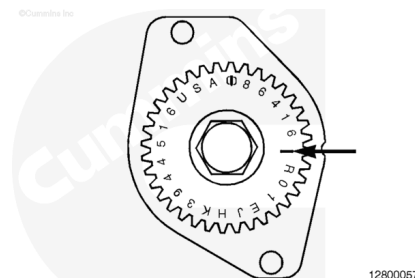
Note : There are two similar marks on the air compressor gear that look like “[I]” and “l”. The timing mark to be used when timing the air compressor to the engine is “l”.



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Cummins® Single-Cylinder AirCompressors

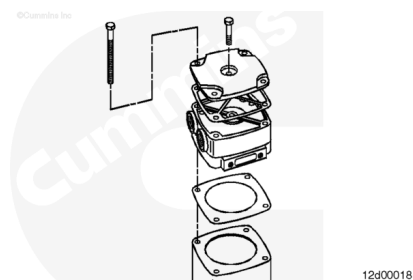
Viewing the compressor from the gear end (with the compressor in a vertical position), rotate the gear so the “l” timing mark is at the 3-o'clock position. The Cummins® single-cylinder air compressor will have a divot on the housing at the 3-o'clock position to aid in timing the compressor. Once the timing mark is at the 3-o'clock position, this will set the compressor at 60-degrees before top dead center of the compressor's compression stroke.



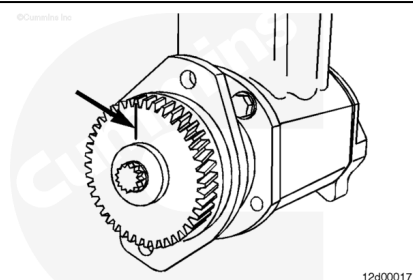
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Air Compressor Timing (for Single-Cylinder Air Compressor Only, Other than Holset® or Cummins®)

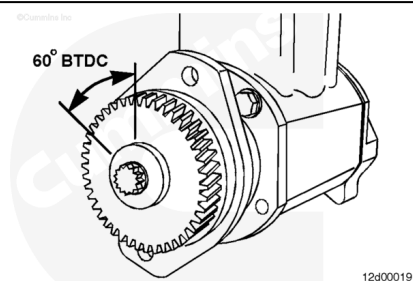
Locate TDC on the compressor crankshaft by removing the unloader valve or head. (Refer to the air compressor manual.) TDC does **not** have to be exact. The system is tolerant of some misalignment.



Use ink or dykem to mark the air compressor gear face at TDC (12-o'clock position when viewed from the front).

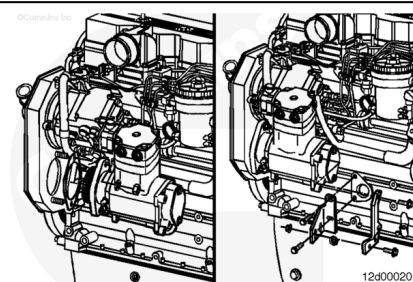


Rotate the compressor TDC mark to 60 degrees, or six teeth on a 36-tooth gear, before TDC. This is approximately 10-o'clock when viewed from the front of the air compressor.



Install the air compressor and new gasket to the gear housing.

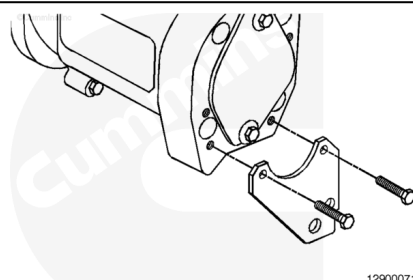
Torque Value: 77 n•m [57 ft-lb]



Install the brace to the air compressor.

Tighten the capscrews.

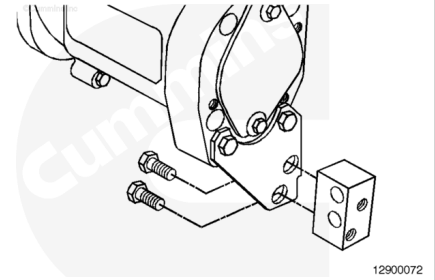
Torque Value: 43 n•m [32 ft-lb]



Install the spacer to brace capscrews and hand-tighten. Make sure there are no gaps between the spacer and the brace, and no gaps between the spacer and the cylinder block.

Tighten the capscrews.

Torque Value: 43 n•m [32 ft-lb]

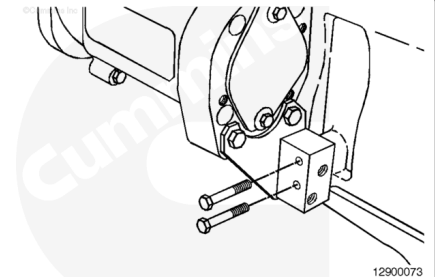


Install the capscrews that connect the spacer to the cylinder block and hand-tighten.

Make sure there are no gaps between the spacer and the brace, and no gaps between the spacer and the cylinder block.

Tighten the capscrews.

Torque Value: 43 n•m [32 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.

- Install the air inlet and air outlet connections to the air compressor.
- Install the air governor or air governor signal line, if necessary. Refer to the OEM service manual.
- Install the air compressor coolant lines.
- Install the air lines.
- Install any attachments to the rear of the air compressor. Refer to the OEM service manual.

- Fill the engine cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine to activate the air compressor and check for leaks.

Last Modified: 26-Aug-2014
